# Current Affairs 29-10-2026

## Credit - Drishti Ias

### Special Executive Magistrates

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**Source: TH**

## Why in News?

A **Special Executive Magistrate (SEM)** court in Delhi granted bail to two activists, rejecting the police's request for extended custody over an alleged breach of peace bonds. The case highlights the **preventive powers, jurisdiction and role of SEMs compared with Judicial Magistrates.**

## Who are Special Executive Magistrates (SEMs)?

- **Appointment:** Under Section 15 of the **Bharatiya Nagarik Suraksha Sanhita (BNSS), 2023** , State Governments may appoint Executive Magistrates or police officers **not below the rank of Superintendent of Police (SP)** or equivalent as SEMs for specified areas, periods or functions.
- **Police as Magistrates:** In commissionerate systems like Delhi, the **Delhi Police Act, 1978** allows the Centre to authorise police officers of the rank of Assistant Commissioner of Police (ACP) and above to exercise the powers of an executive magistrate.
- **Nature of Role:** SEMs belong to the Executive branch. Their primary function is strictly **preventive and administrative** to intervene before a disturbance happens, rather than to punish an individual for an offence that has already been committed.
- **Key Powers of SEMs** The powers of SEMs are largely preventive and are codified under the BNSS, 2023:
  - **Issuing "Peace Bonds" (Section 126, BNSS):** If an SEM suspects a person is likely to **breach public peace,** they can ask the individual to show cause as to why they should not execute a formal bond promising good behaviour for a period of up to one year.
  - **Preventive Inquiry (Section 135, BNSS):** The magistrate can conduct an inquiry and order security during the process, provided they record their reasons in writing.
  - **Detention for Breach (Section 141(1)(b), BNSS):** If an individual breaks the conditions of their peace bond, the SEM can bypass a normal trial and order their immediate arrest and detention for the remaining period of the bond.
  - **Urgent Prohibitory Orders (Section 163, BNSS):** SEMs can issue urgent preventive orders against nuisance or apprehended danger to human life, health, safety, or public tranquillity (e.g., preventing riots or obstruction).
  - **Bail in Preventive Crackdowns:** While they do not handle ordinary criminal bail, SEMs have the authority to accept sureties and grant bail or order temporary police custody during preventive

proceedings.

- **Judicial Review:** The actions and orders of an SEM are entirely subject to **judicial review** and administrative oversight to ensure they are legal and that citizens' rights are not arbitrarily violated.
- **Right to Challenge:** Depending on the nature of the specific order and the legal remedies available, a citizen can challenge an SEM's decision before an appropriate higher court or authority.

# What are the Concerns Regarding the Powers of SEMs?

- **Threat to Personal Liberty:** Preventive arrests without a formal trial raise severe concerns regarding **Article 21** (Right to Life and Personal Liberty) and **Article 22** (Protection against arbitrary arrest and detention) of the Constitution.
- **Mechanical Orders and "Cyclostyled" Proformas:** Courts have repeatedly flagged the indiscriminate use of SEM powers.
- **Procedural Safeguards and Accountability:** In ***Aldanish Rein v. State of NCT of Delhi (2018),*** the Delhi High Court emphasised reasoned orders, periodic detention reviews, affordable sureties, access to legal aid and independent oversight to prevent arbitrary exercise of SEM powers.
- **Colonial Legacy:** The preventive powers of SEMs trace their origins back to **England's *Justices of the Peace Act, 1361* ,** which entered Indian law in 1861 to suppress dissent, retaining a colonial overtone.
- **Conflict of Interest and Natural Justice:** When police officers act as SEMs, the same executive machinery may initiate preventive proceedings and adjudicate them, raising concerns under the principle of Nemo Judex in Causa Sua (no one should be a judge in their own cause) and **Article 50** (separation of judiciary from executive).

## SEMs vs. Judicial Magistrates

| Feature | Special Executive Magistrates (SEMs) | Judicial Magistrates |
|---|---|---|
| **Appointment** | Appointed by the **State Government** . | Courts of Judicial Magistrates (first and second class) are established under **Section 9 of the BNSS** . While the State determines their number and location, the presiding officers are **appointed by the High Court,** not the State government. |
| **Primary Role** | Administrative and preventive (maintaining law and order). | Judicial Magistrates are an **independent part of the judicial system** tasked with dealing with formal criminal cases, hearing regular bail applications, conducting trials, and imposing legal sentences. |
| **Judicial Powers** | Cannot conduct formal criminal trials, convict, or impose sentences. | Presides over criminal courts, handles trials, and sentences offenders. |

## Frequently Asked Questions (FAQs)

**1. What is a Special Executive Magistrate (SEM)?**
A SEM is an executive authority appointed by the government under Section 15 of the BNSS, 2023, to exercise specified magisterial powers for maintaining public order and preventing breaches of peace.

**2. What are the key preventive powers of SEMs under the BNSS, 2023?**
Their powers include requiring security for keeping the peace under Section 126, conducting inquiries under Section 135 and issuing urgent orders under Section 163, subject to applicable legal conditions.

**3. How do Special Executive Magistrates differ from Judicial Magistrates?**
SEMs primarily exercise executive and preventive powers, while Judicial Magistrates handle criminal proceedings, conduct trials within their jurisdiction and exercise judicial powers independently.

**4. How do SEM powers raise concerns regarding fundamental rights?**
Preventive proceedings and detention may affect Article 21 (right to life and personal liberty) and Article 22 (safeguards against arrest and detention), making procedural safeguards and judicial review essential.

**5. Why is the separation of powers relevant to SEMs?**
When police officers exercise magisterial powers, concerns may arise over impartiality and accountability. Article 50 directs the State to separate the judiciary from the executive in public services.

**[Watch Video on YouTube: ▶ https://www.youtube.com/embed/gahGpc2ra8M ]**

## UPSC Civil Services Examination, Previous Year Questions

### *Prelims*

**Q. Consider the following statements: (2023)**

1. According to the Constitution of India, the Central Government has a duty to protect States from internal disturbances.
2. The Constitution of India exempts the States from providing legal counsel to a person being held for preventive detention.
3. According to the Prevention of Terrorism Act, 2002, confession of the accused before the police cannot be used as evidence.

**How many of the above statements are correct?**

(a) Only one

(b) Only two

(c) All three

(d) None

**Ans: B**

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# India's Emerging Space Technologies

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**Source: PIB**

## Why in News?

The **Indian Space Research Organisation (ISRO)** is advancing **next-generation launch vehicles** , **reusable rockets** , **indigenous propulsion systems** and **in-space docking technologies** to strengthen India's space capabilities and support future missions such as **Gaganyaan** and **Bharatiya Antariksh Station (BAS) .**

## What are the Key Developments in India's Space Sector?

- **Evolution of India's Space Programme:** India's organised space activities began with the establishment of **INCOSPAR in 1962** , followed by the **first sounding rocket launch from Thumba, Kerala, in 1963** and the **establishment of ISRO in 1969.**

- **Technological Self-Reliance:** India is developing indigenous capabilities in **reusable launch vehicles, advanced propulsion, space docking, satellite navigation and Earth observation** to reduce dependence on foreign technologies.
- **Human Spaceflight Ambitions:** The **Gaganyaan Programme** aims to demonstrate indigenous human spaceflight capability, while the planned **Bharatiya Antariksh Station** will support future long-duration human space missions.
- **Private-Sector Participation:** Growing involvement of space startups and private industries is strengthening indigenous manufacturing, technology development and commercial space services.
- **Space Economy Expansion:** The Department of Space targets a **USD 44-billion space economy by 2033,** supported by indigenous technologies, commercial missions and private investment.
- **Industry and Startup Development: NewSpace India Limited (NSIL) has signed 118 Technology Transfer Agreements** covering 83 ISRO-developed technologies, strengthening domestic industrial capabilities.
  - Cumulative investment in Indian space startups reached approximately **USD 618.5 million by March 2026.**
- **Space Sector Reforms:** The **Indian Space Policy, 2023 , and IN-SPACe** have enabled greater private participation in satellite manufacturing, launch services and other commercial space activities .

# What are the Key Emerging Space Technologies Being Developed by India?

## Next Generation Launch Vehicle (NGLV)

- **About:** The Next Generation Launch Vehicle (NGLV) is an **advanced launch system being developed by ISRO to enable heavy-lift,** partially reusable and cost-effective access to space.
- **Government Approval:** In **September 2024, the Union Cabinet approved its development with an allocation of ₹8,240 crore** , including three developmental flights.
- **Payload Capacity:**
  - Designed to carry up to 30 tonnes to **Low Earth Orbit (LEO),** compared with approximately 10 tonnes for LVM3.
  - A partially reusable configuration is being developed with a 14-tonne LEO payload capacity.
- **Applications:** Will support **Bharatiya Antariksh Station, human spaceflight, lunar exploration, interplanetary missions** and satellite constellations.

## India’s Traditional Launch Vehicles

- **PSLV (Polar Satellite Launch Vehicle) :** ISRO’s **workhorse launch vehicle** , capable of carrying up to **1,750 kg to a 600-km Sun-Synchronous Polar Orbit (SSPO)** and up to **1,425 kg to Sub-Geosynchronous Transfer Orbit (Sub-GTO)** , depending on mission configuration.
- **GSLV (Geosynchronous Satellite Launch Vehicle) :** Equipped with an **indigenous Cryogenic Upper Stage** , GSLV can launch communication satellites of up to the **2-tonne class** .
- **LVM3 (Launch Vehicle Mark-3) :** India’s **heavy-lift launch vehicle** , capable of placing communication satellites of the **4-tonne class into GTO** and payloads of the **10-tonne class into LEO** .
- **SSLV (Small Satellite Launch Vehicle) :** A **three-stage, approximately 120-tonne launch vehicle** designed for small satellites, capable of placing payloads of up to **500 kg into a 500-km planar orbit** .

## Key Terms

- **Sun-Synchronous Orbit (SSO):** A near-polar orbit in which a satellite passes over a given area at approximately the **same local solar time** , allowing observations under similar lighting conditions. The revisit period varies depending on the satellite and its orbital parameters.
- **Geostationary Transfer Orbit (GTO):** An **elliptical intermediate orbit** used to transfer a satellite into geostationary orbit (GEO). Its perigee is generally a few hundred kilometres above Earth, while its apogee is close to **35,786 km** . Near apogee, the satellite uses its **apogee motor** to raise its perigee and circularise the orbit before entering GEO.
- **Cryogenic Upper Stage:** The final stage of a launch vehicle that uses **liquid hydrogen ($LH_2$) as fuel and liquid oxygen (LOX) as oxidiser** . These propellants provide high efficiency and enable the launch vehicle to place heavier payloads into higher orbits.
  - ISRO’s indigenous cryogenic stage replaced the earlier Russian-supplied stage used in GSLV missions, strengthening India’s **self-reliance in advanced launch technology** .

# Reusable Launch Vehicle (RLV) Technology

- **About:** Reusable launch vehicles are **designed to recover and reuse major rocket components,** reducing launch costs and improving operational efficiency.
- **Pushpak RLV Demonstrator** : ISRO successfully conducted three **Reusable Launch Vehicle Landing Experiments (RLV-LEX)** in 2023 and 2024.
  - The **LEX-03 experiment in June 2024** demonstrated autonomous runway landing under challenging wind and cross-range conditions.
- **Significance:** Advances indigenous capabilities in autonomous navigation, controlled atmospheric re-entry and precision landing.

## Scramjet and Air-Breathing Propulsion

- **Scramjet Technology:** A **Supersonic Combustion Ramjet (Scramjet)** is an air-breathing engine that uses atmospheric oxygen for combustion, reducing the need to carry onboard oxidiser during atmospheric flight.
- **Working Principle:** Operates at hypersonic speeds (above Mach 5), with combustion occurring in supersonic airflow.
- **ISRO Achievements:**
  - Successfully flight-tested a scramjet engine in August 2016, demonstrating hypersonic combustion at approximately Mach 6.
  - Conducted an **RH-560 sounding rocket** experiment in July 2024 to evaluate air-breathing propulsion systems.
- **Significance:** Offers potential for reducing propulsion-system mass and improving the efficiency of future space transportation systems.

# Electric and Green Propulsion Technologies

- **Electric Propulsion:** Uses electrical energy to accelerate propellant and produce thrust, enabling efficient satellite orbit raising and station-keeping with lower propellant consumption.
  - In **March 2025, ISRO completed a 1,000-hour life test of its indigenous 300-millinewton stationary plasma thruster** .
  - The technology offers a specific impulse at least six times greater than conventional chemical propulsion systems, and is planned for demonstration aboard TDS-01.
- **Green Propulsion:** Focuses on **developing alternative propellants and propulsion technologies** to reduce environmental and operational concerns.
  - **ISROSENE:** An indigenous rocket-grade kerosene being developed for liquid oxygen–kerosene semicryogenic propulsion systems.
  - **Bellatrix Aerospace:** Developed a 1-newton-class green monopropellant thruster under **DRDO's Technology Development Fund** , with in-orbit functionality demonstrated in January 2024.

## Radioisotope Heating and Power Technologies

- **Radioisotope Heater Units (RHUs):** Use heat released by radioactive decay to maintain spacecraft components at suitable operating temperatures in extremely cold environments.
- **Radioisotope Thermoelectric Generators (RTGs):** Convert heat from radioactive decay into electrical energy, supporting long-duration missions where solar power may be inadequate.
- **Indian Initiative:** ISRO and the **Department of Atomic Energy (DAE)** are collaborating on RHU and RTG requirements for potential future Chandrayaan and Mars missions.

# Space Docking and In-Orbit Servicing

- **SpaDeX Mission:** ISRO's Space Docking Experiment (SpaDeX) demonstrated the capability to rendezvous, dock and undock two spacecraft in orbit.
  - **First successful docking occurred on 16th January 2025,** making **India the fourth country** to demonstrate space docking after the United States, Russia and China.
  - Successfully demonstrated undocking in March 2025, followed by a **second autonomous docking and bidirectional power transfer in April 2025.**
- **In-Orbit Servicing:** Future applications could include **satellite refuelling, maintenance, repair and upgrades, potentially extending spacecraft lifespans** .
- **Strategic Importance:** Docking technology is essential for **lunar sample-return missions, multi-launch operations and assembly of Bharatiya Antariksh Station.**

# Indigenous Satellite Navigation and Communication

- **Navigation with Indian Constellation (NavIC):** India's indigenous regional navigation system provides positioning, navigation and timing services over India and approximately 1,500 km beyond the Indian mainland.
  - **Offers Standard Positioning Service (SPS)** for civilian users and encrypted **Restricted Service (RS)** for authorised users.
- **Advanced Communication Satellites:**
  - **GSAT-24 (2022):** Supports Direct-to-Home (DTH) television broadcasting.
  - **GSAT-N2/GSAT-20 (2024):** A high-throughput communication satellite with approximately 48 Gbps capacity, supporting broadband and in-flight connectivity.
  - **CMS-03/GSAT-7R (2025):** A dedicated communication satellite for the Indian Navy, strengthening maritime communications and national security.

# Advanced Earth Observation Technologies

- **NASA–ISRO Synthetic Aperture Radar (NISAR):** Launched on 30 July 2025, it is a joint Earth observation mission using **NASA's L-band and ISRO's S-band radar frequencies.**
  - Supports all-weather, day-and-night observations and monitors ground deformation, glaciers, vegetation, soil moisture and natural hazards.
  - Designed to **observe Earth's land and ice-covered surfaces** over a 12-day repeat cycle.
- **Hyperspectral Imaging Satellite (HysIS):** Launched in **November 2018,** it observes Earth across numerous narrow spectral bands.
  - Supports **agriculture, forestry, mineral exploration, soil analysis and water-resource monitoring** by identifying material-specific spectral characteristics.




# Conclusion

India's emerging space technologies mark a shift towards a **self-reliant, innovation-driven and commercially competitive space ecosystem.** Sustained investment in research, greater private-sector participation and stronger industry–academia collaboration will be crucial for translating technological advancements into economic opportunities. These efforts will strengthen India’s strategic autonomy, scientific leadership and position in the global space economy.

***Drishti Mains Practice Question***

Emerging space technologies are transforming India's capabilities in launch systems, satellite operations and deep-space exploration. Discuss.

# Frequently Asked Questions (FAQs)

**Q1. What is India's Next Generation Launch Vehicle (NGLV)?**
NGLV is an advanced launch vehicle under development by ISRO, designed to carry up to 30 tonnes to Low Earth Orbit, with a partially reusable configuration being developed for cost-effective space access.

**Q2. What is the SpaDeX Mission?**
SpaDeX (Space Docking Experiment) is an ISRO mission that demonstrated autonomous docking, undocking and power transfer between spacecraft in orbit, making India the fourth country to achieve space docking.

**Q3. What is the difference between electric and chemical propulsion?**
Electric propulsion uses electricity to accelerate propellant and generally provides higher propellant efficiency but lower thrust, whereas chemical propulsion produces relatively high thrust through chemical reactions.

**Q4. What is NavIC?**
Navigation with Indian Constellation (NavIC) is India's indigenous regional satellite navigation system, providing positioning, navigation and timing services over India and approximately 1,500 km beyond the Indian mainland.

**Q5. What is the NISAR Mission?**
NISAR (NASA–ISRO Synthetic Aperture Radar) is a joint Earth observation satellite launched in July 2025, using L-band and S-band radar to monitor changes in Earth's surface, including glaciers, vegetation and ground deformation.

**[Watch Video on YouTube: ▶ https://www.youtube.com/embed/aQIjtwZrq9U ]**

## UPSC Civil Services Examination, Previous Year Question (PYQ)

### *Prelims*

**Q. With reference to India's satellite launch vehicles, consider the following statements: (2018)**

1. PSLVs launch the satellites useful for Earth resources monitoring whereas GSLVs are designed mainly to launch communication satellites.
2. Satellites launched by PSLV appear to remain permanently fixed in the same position in the sky, as viewed from a particular location on Earth.
3. GSLV Mk III is a four-staged launch vehicle with the first and third stages using solid rocket motors; and the second and fourth stages using liquid rocket engines.

**Which of the statements given above is/are correct?**

(a) 1 only

(b) 2 and 3

(c) 1 and 2

(d) 3 only

**Ans: (a)**

**Q. Consider the following statements: (2016)**

1. The Mangalyaan launched by ISRO
2. is also called the Mars Orbiter Mission
3. made India the second country to have a spacecraft orbit the Mars after USA
4. made India the only country to be successful in making its spacecraft orbit the Mars in its very first attempt

**Which of the statements given above is/are correct?**

(a) 1 only

(b) 2 and 3 only

(c) 1 and 3 only

(d) 1, 2 and 3

**Ans: (c)**

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### *Mains*

**Q.** What is the main task of India's third mood mission which could not be achieved in its earlier mission? List the countries that have achieved this task. Introduce the subsystems in the spacecraft launched and explain the role of the 'Virtual Launch Control Centre' at the Vikram Sarabhai Space Centre which contributed to the successful launch from Sriharikota. **(2023)**

**Q.** What is India's plan to have its own space station and how will it benefit our space programme? **(2019)**

**Q.** India has achieved remarkable successes in unmanned space missions including the Chandrayaan and Mars Orbiter Mission, but has not ventured into manned space mission, both in terms of technology and logistics? Explain critically. **(2017)**

**Q.** Discuss India's achievements in the field of Space Science and Technology. How the application of this technology helped India in its socio-economic development? **(2016)**

# Defence Acquisition Council Clears Procurement Proposals Worth ₹2.12 Lakh Crore

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**Source: TH**

Recently, the **Defence Acquisition Council (DAC)** , chaired by Defence Minister, accorded **Acceptance of Necessity (AoN)** for defence procurement proposals worth approximately ₹2.12 lakh crore, with around **82% of the proposed procurements expected to be sourced from Indian industry.**

- **Acceptance of Necessity (AoN):** It is an **initial approval** in the defence acquisition process indicating the need for a proposed procurement; it **does not mean the contracts have been awarded.**
- **Major acquisition proposals:**
  - **Indian Army:** Proposed acquisitions include **High Mobility Vehicles, self-propelled tracked gun systems, Light Multi-Role Helicopters , Very Short-Range Air Defence Systems (VSHORADS) , counter-drone systems** and **long-range drones.**
    - **Counter-Unmanned Aerial Systems (C-UAS) and surveillance drones** will improve protection against aerial threats and strengthen reconnaissance and targeting.
  - **Indian Navy: Active Offboard Decoy Systems (AODS)** will strengthen protection against anti-ship missile threats, alongside other naval platform acquisitions.
  - **Indian Air Force:** Proposed acquisitions include **helicopters, air defence systems, weapons and Mobile X-Band Doppler Weather Radar Systems** for localised weather surveillance in forward areas.
- **Domestic Manufacturing** : The proposed procurement is expected to support **indigenous defence production, technology development and the domestic defence industrial ecosystem.**
  - India's annual defence production reached a record **₹1.78 lakh crore in FY 2025-26,** reflecting **15.6% growth** over the previous financial year.
- **Significance** : Greater domestic sourcing can **reduce import dependence, strengthen military preparedness** and support India’s goal of **Aatmanirbhar Bharat in defence.**

**Read more:** **DAC Granted Acceptance of Necessity for Defence Acquisition Proposals**

# Amendments to Central Motor Vehicle Rules for E100 Fuel Usage

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**Source: TH**

The Ministry of Road Transport and Highways (MoRTH) has amended the Central Motor Vehicles Rules (CMVR), 1989, to permit the use of **E100 (up to 100% ethanol)** as an automotive fuel in specially designed and certified **flex-fuel vehicles (FFVs).**

- **Statutory Amendments:** The government has modified **Rule 115 (dealing with emission standards)** and **Rule 115E (dealing with emission testing)** of the CMVR, 1989, to include E100 and E85 in the official fuel specification tables.
- **Target Vehicles:** The newly approved **E85 (85% ethanol)** and **E100 (100% ethanol)** fuels are strictly meant to be used in vehicles designed by **Original Equipment Manufacturers (OEMs) as dedicated flex-fuel vehicles** .
- **No Mandatory Transition:** The Ministry clarified that the notification **does not mandate existing vehicle owners to switch to E85 or modify their current engines** , easing public concerns over the transition.
- **Compatibility Distinction:** A vehicle engineered to run on **E10 or E20 (10% or 20% ethanol blending) is not automatically suitable for E85 or E100** . Motorists must rely on the **fuel compatibility sticker** and manufacturer guidelines.
- **Flex-Fuel Vehicles (FFVs):** These vehicles can operate on petrol and varying ethanol blends, potentially up to E100, using a single fuel tank. Their engine control systems automatically adjust to the fuel composition within the range certified by the manufacturer.
- **Ethanol (Ethyl Alcohol):** It is a renewable biofuel produced naturally by the fermentation of sugars by yeasts or via **petrochemical processes.**
  - In India, it is primarily derived from **sugarcane molasses, maize, and surplus food grains.**
- **Ethanol Blended Petrol (EBP) Programme** : Under the **National Policy on Biofuels, 2018** , the **Ethanol Blended Petrol (EBP) Programme** (launched in 2003) aims to curb crude oil imports, cut greenhouse gas emissions, and boost farmers' incomes by expanding permissible raw materials to include sugarcane juice, sugar beet, corn, and damaged grains.
  - The initiative reached a historic milestone by achieving its **20% ethanol blending target (E20) in 2025** , five years ahead of the original 2030 deadline.
  - Building on this success, as of **April 2026** , it is mandatory that all petrol sold across India contains 20% ethanol and meets a minimum **Research Octane Number (RON) of 95** .

**Read more:** **India’s Ethanol Blending Programme and the Path to E100**

# TDB Supports Indigenous Green Propulsion Technology

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**Source: DD**

The **Technology Development Board (TDB),** under **the Department of Science and Technology (DST),** signed an agreement with **Manastu Space Technologies Pvt. Ltd.,** Navi Mumbai, to provide ₹116 crore under the **Research, Development and Innovation (RDI) Fund** for developing indigenous green propulsion systems for satellites and launch vehicles.

- **Project Cost:** Estimated at ₹277.63 crore, with financial assistance provided through **Optionally Convertible Debt (OCD)** , a financing instrument that can be converted into equity under specified conditions.
- **Scalable Architecture:** Designed to deliver thrust ranging from 5 millinewtons (mN) to 300 newtons (N), supporting applications from small satellites to higher-thrust systems for launch vehicles.
- **Key Components:** Includes **propulsion engines, propellant feed systems and an ultra-high-temperature ceramic catalyst** for reliable performance under demanding conditions.
- **Technology Readiness Level (TRL):** The project aims to advance the technology from **TRL 6 (technology demonstrated in a relevant environment) to TRL 9 (actual system proven in operational use)** .
  - The propulsion technology has already been demonstrated in space, with further **engineering, system integration, testing and validation** planned to enable wider operational deployment.
- **Green Propulsion Technology:** Uses a hydrogen peroxide-based propellant with additives, supported by an integrated engine and catalyst system to improve operational safety, performance and cost efficiency.
- **Technology Development Board (TDB):** A statutory body constituted under the **Technology Development Board Act, 1995, functioning under the Department of Science and Technology, Ministry of Science and Technology.**
  - **It provides financial assistance for the development and commercialisation of indigenous technologies** and adaptation of imported technologies for domestic applications.
- **RDI Fund:** Approved in **July 2025** with an outlay of **₹1 lakh crore over six years** , the scheme aims to promote **private-sector research, development and innovation** in strategic and emerging technologies.
  - It operates through a **two-tier funding mechanism under the Anusandhan National Research Foundation (ANRF)** , with TDB serving as a **Second-Level Fund Manager** to support high-risk, technology-intensive projects.

**Read More:** **Green Propulsion System**